Econometrics III: Applied Methods — NovaSBE, 2026

Problem Set 4 Solutions: Regression Discontinuity Design

Exercise A: Assessing an RDD

1.1 *(0.5 pts)* Write down the continuity assumption required for identification in this RDD. What must be true about potential outcomes at the cutoff?

Solution:

The key identifying assumption for RDD is the **continuity assumption**:

$$\lim_{x \downarrow c} \mathbb{E}[Y_i(0)|X_i = x] = \lim_{x \uparrow c} \mathbb{E}[Y_i(0)|X_i = x]$$

$$\lim_{x \downarrow c} \mathbb{E}[Y_i(1)|X_i = x] = \lim_{x \uparrow c} \mathbb{E}[Y_i(1)|X_i = x]$$

In words: The expected potential outcomes under treatment and under control must be *continuous* at the cutoff.

What this means for the politician study:

- Constituencies just above and just below the vote margin cutoff must be similar in all relevant characteristics that affect public goods provision
- The only thing that should "jump" at the cutoff is whether a dishonest vs. honest politician wins
- If other factors also jump discontinuously at the cutoff, the estimate captures a mixture of effects

1.2 *(1 pt)* Consider the definition of "dishonest": a politician whose income exceeds the tax threshold but who did not pay taxes. What does this definition imply about the relationship between dishonesty status and politician income?

Solution:

The definition of "dishonest" **mechanically creates a correlation with politician income**:

- To be classified as "dishonest," a politician must have income *above* the tax threshold
- Politicians with income *below* the threshold are automatically "honest" regardless of whether they would evade if richer
- Therefore: $\text{Income}_{dishonest} > \text{Income}_{honest}$ **by construction**

Implications:

1. "Dishonest" politicians are necessarily richer than the tax threshold
2. "Honest" politicians include both: (a) those below threshold who might evade if richer, and (b) those above threshold who comply

3. Any comparison of dishonest vs. honest is *also* a comparison of richer vs. poorer politicians

This is a **bundled treatment** problem: "dishonesty" is inseparable from "high income" in this classification.

1.3 *(1.5 pts)* Based on the two tables above, assess the validity of this RDD. Explain. Suppose the true causal effect of dishonesty on public goods provision is zero. Could the paper's findings still arise? Under what conditions?

Solution:

Assessment of validity:

Table 1 (Constituency characteristics): Most predetermined constituency characteristics are balanced at the cutoff. This suggests that constituencies where dishonest politicians barely won are similar to those where honest politicians barely won.

Table 2 (Politician characteristics): However, dishonest and honest politicians differ systematically:

- Dishonest politicians have significantly higher income (960K vs. 607K BDT)
- Dishonest politicians have significantly higher criminal records (33.8% vs. 23.0%)

The problem: The RDD compares constituencies that narrowly elected a dishonest (rich, potentially criminal) politician vs. an honest (poorer, less criminal) politician. Multiple politician characteristics jump at the cutoff, not just "honesty."

Could findings arise with zero effect of dishonesty?

Yes! If the true causal effect of dishonesty is zero but:

1. Richer politicians provide fewer public goods (perhaps because they're less connected to constituents' needs)
2. Politicians with criminal records provide fewer public goods (perhaps due to diversion of resources)
3. Other unobserved politician characteristics correlated with "dishonesty" affect outcomes

The paper's estimate would capture these effects, not the effect of dishonesty per se.

Conclusion: The RDD is valid for identifying the effect of *electing the type of politician who is classified as dishonest* (rich, non-tax-paying, higher criminal record). It is **not** valid for identifying the causal effect of "dishonesty" isolated from these other characteristics.

Exercise B: RDD Estimation in Practice

See accompanying R code in `pset4_rdd_solutions.R`.

B.1 Graphical Analysis

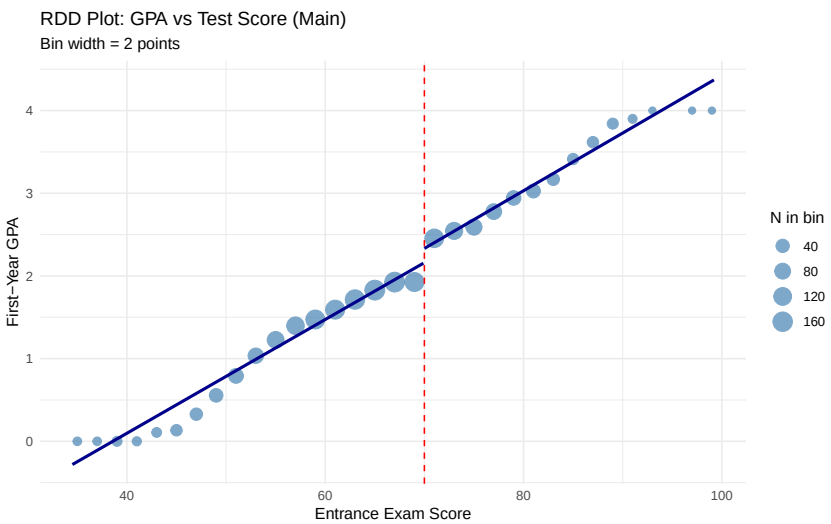
1.1–1.2 *(1 pt)* Create RDD plots with different bin widths. Comment on the trade-off.

Solution:

1.1 Main RDD Plot:

The RDD plot shows a clear discontinuity at the cutoff of 70. Mean GPA jumps visibly at the threshold.

R Output:



Stata Output:

1.2 Bin Width Comparison:

Bin Width	Characteristics
1 point	Noisy—few observations per bin, but shows local variation in detail
5 points	Good balance between signal and noise; bins have reasonable sample sizes
10 points	Smooth appearance, but may obscure local patterns near the cutoff

Trade-off:

- **Narrow bins:** Show more detail but are noisier (high variance, low bias)
- **Wide bins:** Smoother but may hide relevant patterns (low variance, high bias)

Recommendation: For presentation, bin width of 2–5 is typically appropriate. The goal is to show the discontinuity clearly without misleading smoothness.

B.2 Local Linear Estimation

2.1–2.3 *(1.5 pts)* Estimate RDD by hand and with rdrobust. Compare.

Solution:

2.1 By-hand estimate (bandwidth $h = 10$):
Regression: $Y_i = \alpha + \tau D_i + \beta_1(X_i - c) + \beta_2 D_i \cdot (X_i - c) + \varepsilon_i$

	Estimate	Std. Error	t-value	p-value
$\hat{\tau}$ (RDD effect)	$\boxed{0.358}$	0.033	10.88	< 0.001

N observations used: 1,254
2.2 rdrobust estimate:

Quantity	Value
Point estimate	$\boxed{0.462}$
Robust SE	0.050
MSE-optimal bandwidth	5.92
Effective N (left)	473
Effective N (right)	327

2.3 Comparison:
The estimates differ (0.358 vs. 0.462) due to:

1. **Different bandwidths:** By-hand used $h = 10$; rdrobust used MSE-optimal $h = 5.66$
2. **Kernel weighting:** rdrobust uses triangular kernel (observations near cutoff get more weight)
3. **Bias correction:** rdrobust applies bias correction for improved inference

Both estimates are close to the true effect ($\tau = 0.4$). The rdrobust estimate is slightly higher because the narrower bandwidth focuses on observations closest to the cutoff.

B.3 Bandwidth Sensitivity

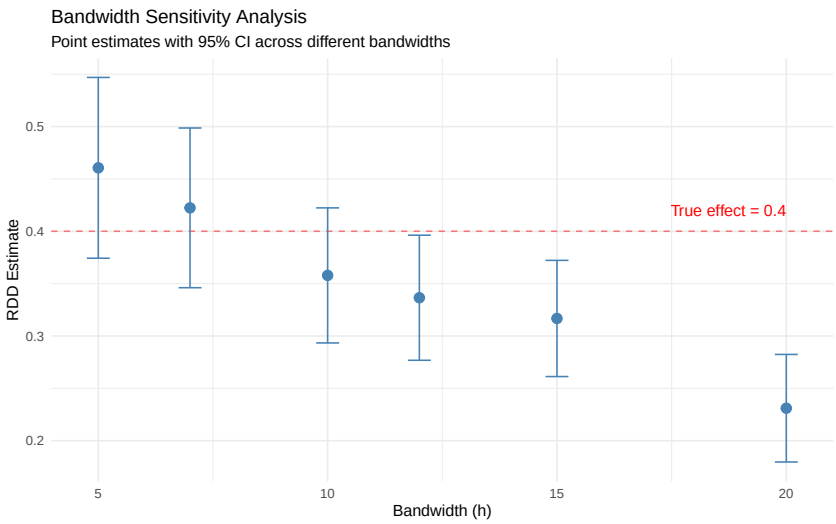
3.1–3.3 *(1 pt)* Bandwidth sensitivity analysis.

Solution:

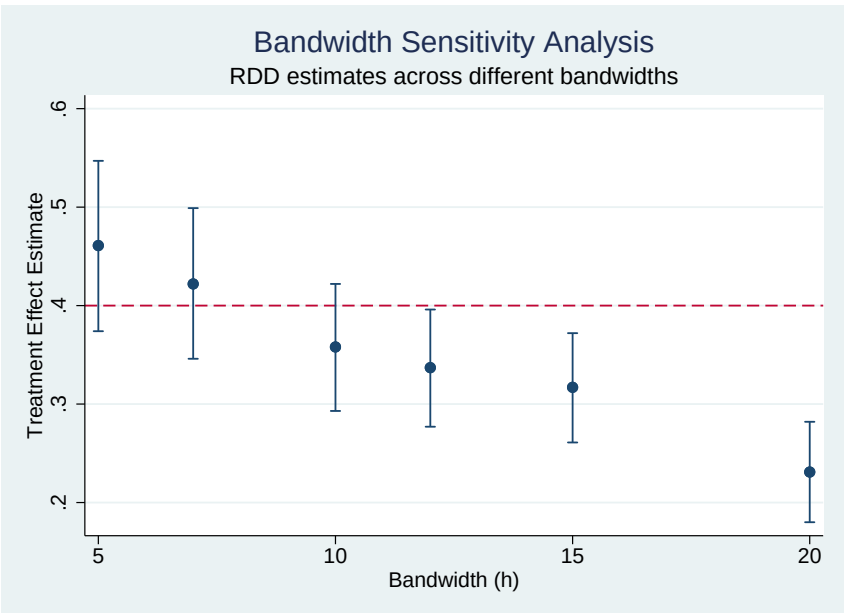
3.1 Bandwidth Sensitivity Table:

Bandwidth	Estimate	Std. Error	N (left)	N (right)
5	0.461	0.044	407	291
7	0.422	0.039	560	374
10	0.358	0.033	788	466
12	0.337	0.030	933	510
15	0.317	0.028	1101	552
20	0.231	0.026	1280	592

R Output:



Stata Output:



3.2–3.3 Commentary:

- Estimates **vary substantially** with bandwidth (range: 0.23–0.46)
- The data has asymmetric curvature on each side of the cutoff, so wider bandwidths pick up this curvature and introduce bias
- Confidence intervals **widen** as bandwidth narrows (fewer observations)
- Narrower bandwidths (closer to the cutoff) give estimates closer to the true effect (0.4)

Bias-variance trade-off:

- **Narrow bandwidth:** Lower bias (more local), higher variance (fewer obs)
- **Wide bandwidth:** Higher bias (includes obs far from cutoff), lower variance (more obs)

The MSE-optimal bandwidth balances these concerns.

B.4 High-Order Polynomials

4.1–4.4 *(1.5 pts)* High-order polynomial analysis and Gelman-Imbens critique.

Solution:

4.1–4.2 Polynomial Order Comparison:

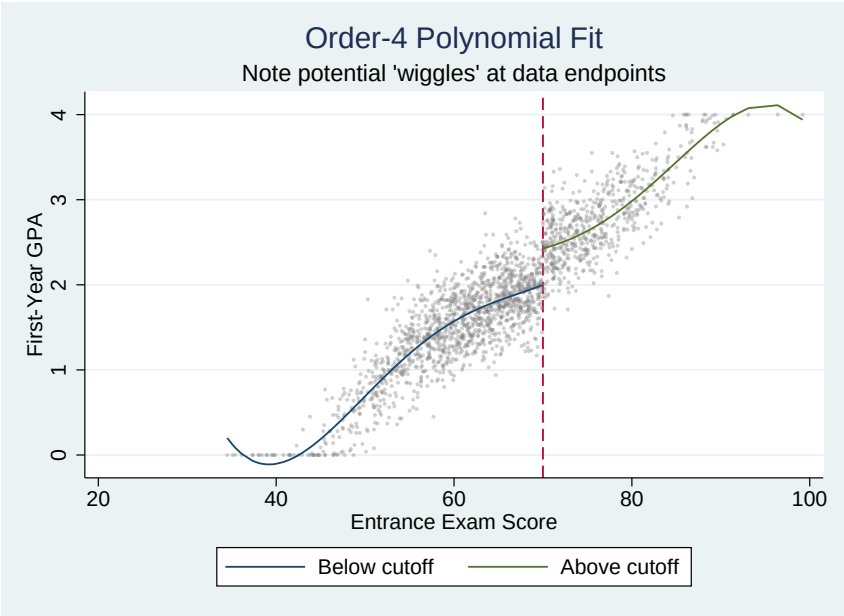
Order	Estimate	Std. Error
1	0.172	0.025
2	0.339	0.033
3	0.531	0.040
4	0.427	0.049

Estimates vary **dramatically** with polynomial order (range: 0.17–0.53), illustrating sensitivity to specification. Standard errors also grow with order.

4.3 Order-4 Polynomial Fit:

R Output:

Stata Output:



The quartic polynomial exhibits clear "wiggles" at the endpoints where data is sparse—note how the fitted line dips below zero for low scores (an impossible GPA!) and curves back down at high scores.

4.4 Gelman-Imbens Critique:

Gelman and Imbens (2019) recommend **against** high-order polynomials for two main reasons:

1. **Sensitivity at boundaries:** High-order polynomials are driven by observations far from the cutoff. The fitted values at the cutoff can be highly sensitive to observations at the data's extremes.

2. **Noisy inference:** Higher-order terms add parameters that are estimated imprecisely, inflating standard errors and making estimates unstable across specifications.

Recommendation: Use local linear or local quadratic regression with optimal bandwidth selection (via rdrobust) rather than global polynomial fits.

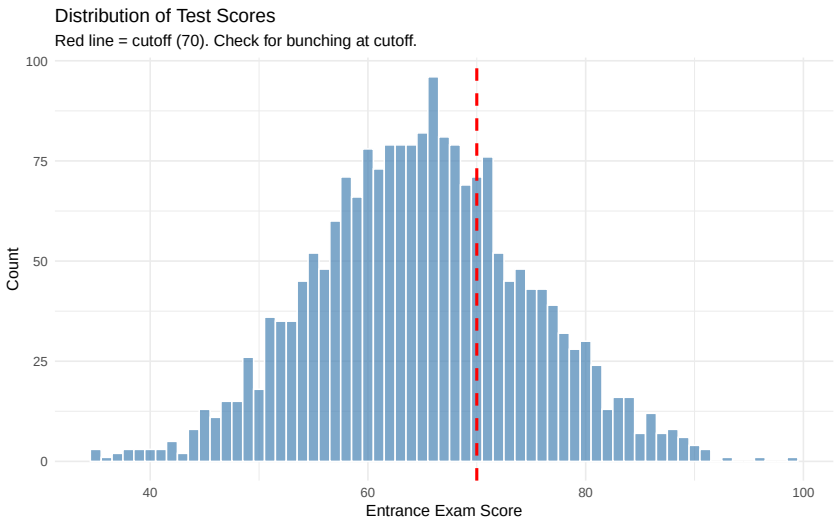
Exercise C: Diagnostics and Robustness

C.1 Testing for Manipulation

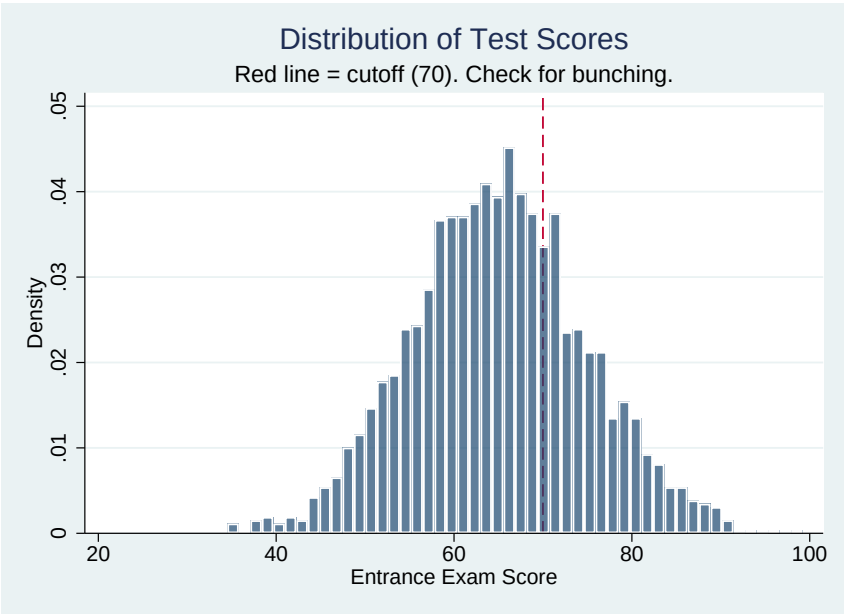
1.1–1.3 *(1 pt)* McCrary density test and interpretation.

Solution:

1.1 Histogram:
R Output:



Stata Output:



Visual inspection shows no obvious bunching at the cutoff. The distribution appears smooth across the threshold.

1.2 McCrary Density Test (rddensity):

Test statistic	1.44
p-value	0.150

Conclusion: We fail to reject the null of no density discontinuity ($p = 0.15 > 0.05$). There is no statistical evidence of manipulation.

1.3 Interpretation if bunching were found:

If students could manipulate their scores:

- **Bunching above cutoff:** High-ability students inflate scores to get scholarship. The treatment group would contain students who "should" be below the cutoff but manipulated their way above.
- **Bias direction:** This would likely bias the estimate **upward**—the scholarship recipients include high-motivation students who would have done well anyway.
- **Validity:** RDD assumes units cannot precisely manipulate the running variable. Manipulation violates this assumption and invalidates the design.

C.2 Covariate Balance

2.1–2.3 *(1 pt)* Test for discontinuities in predetermined covariates.

Solution:

2.1 Covariate Balance Tests:

Covariate	Estimate	Robust SE	p-value
Family Income	−\$2,054	\$2,497	0.256
High School GPA	−0.001	0.046	0.880

Neither covariate shows a significant discontinuity at the cutoff.

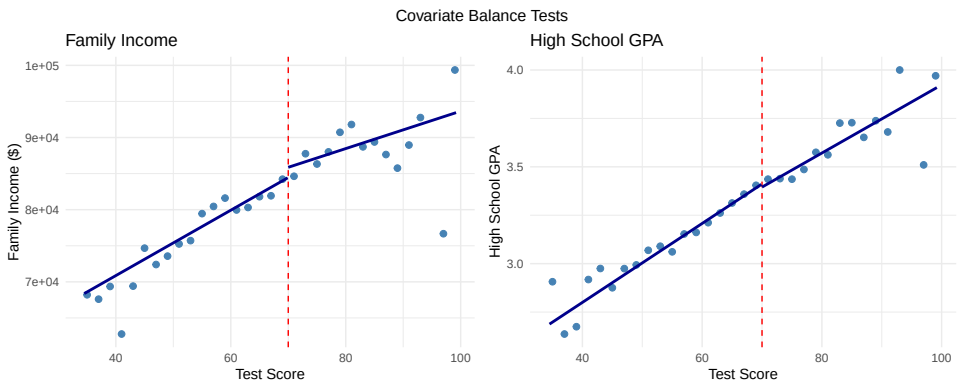
2.2 Why covariate balance matters:

- Predetermined characteristics should be **continuous** at the cutoff—they were determined before the test score
- A discontinuity would suggest either **sorting/manipulation** or **confounding**
- Balance validates the "as-if random" assignment assumption near the cutoff

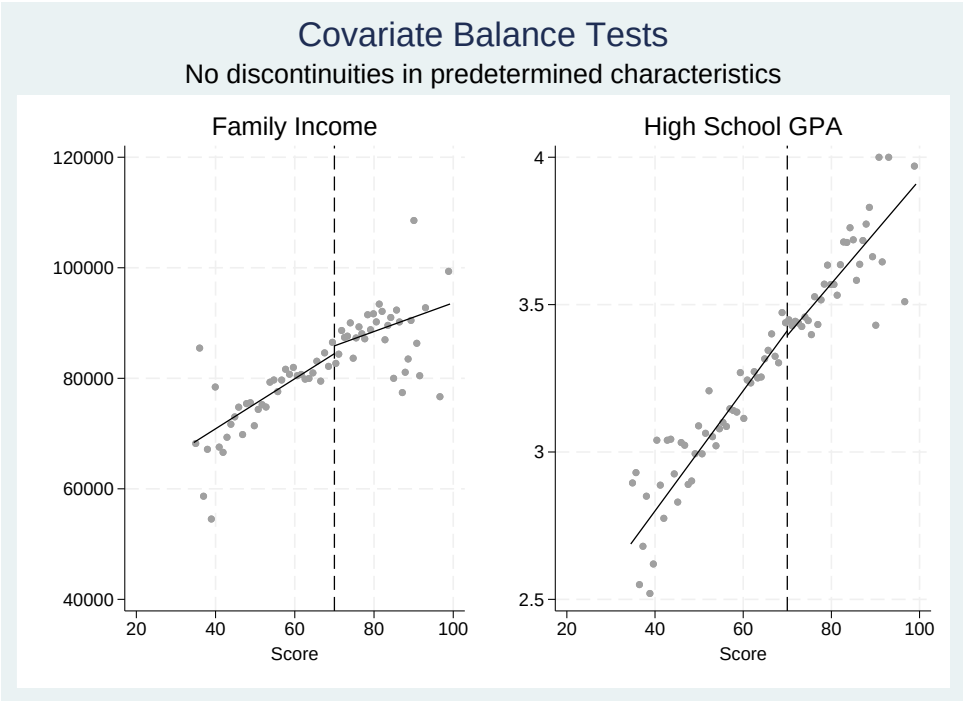
If we found a significant discontinuity, we would conclude that something other than the scholarship is changing at the cutoff, threatening the validity of the causal interpretation.

2.3 Covariate Balance Plots:

R Output:



Stata Output:



Both covariates are continuous through the cutoff, supporting RDD validity.

C.3 Donut Hole Specification

3.1–3.3 *(1 pt)* Donut hole RDD.

Solution:

3.1 Donut Hole Estimate:
Excluding observations with scores in $[68, 72]$:

	Full Sample	Donut Hole
Estimate	0.462	0.506
Robust SE	0.050	0.174

The estimates are similar, though the donut hole estimate has a much larger standard error due to losing observations closest to the cutoff.

3.2 Reasons to use donut hole:

1. **Manipulation:** If students can manipulate scores near the cutoff (e.g., retaking tests, grade bumping), those observations may be "sorted" and unrepresentative. Excluding them removes potentially problematic units.

2. **Measurement error:** Test scores near thresholds may be subject to rounding, grading discretion, or reporting conventions. If scores of 69.5 are sometimes rounded to 70, observations at the exact cutoff may be unreliable.

3.3 Downsides:

- **Loses precision:** Fewer observations, especially the most informative ones near the cutoff

- **RDD is fundamentally local:** We *want* observations near the cutoff for identification
- **Arbitrary:** The donut size (2 points) is a researcher choice that can affect results

Best practice: Show both standard and donut-hole estimates. If they differ substantially, investigate why.

C.4 Asymmetric Bandwidths

4.1–4.3 *(1 pt)* Asymmetric bandwidth analysis.

Solution:

4.1 Distribution of running variable:

N below cutoff	1,401
N above cutoff	599
Ratio (below/above)	2.34

The distribution is asymmetric—roughly 2.3 times as many observations below the cutoff. This is because test scores are drawn from a normal distribution and the cutoff (70) is relatively high.

4.2 Intuition for asymmetric bandwidths:

With more observations below the cutoff:

- We can estimate the conditional expectation function **precisely** on the left with a narrow bandwidth
- On the right, we have fewer observations, so we need a **wider** bandwidth to achieve similar precision

Prediction: The MSE-optimal bandwidth should be **wider on the right** (above cutoff) where data is sparser.

4.3 rdrobust with asymmetric bandwidths:

Using `bwselect = "msetwo"`:

Bandwidth (left/below cutoff)	5.01
Bandwidth (right/above cutoff)	6.07

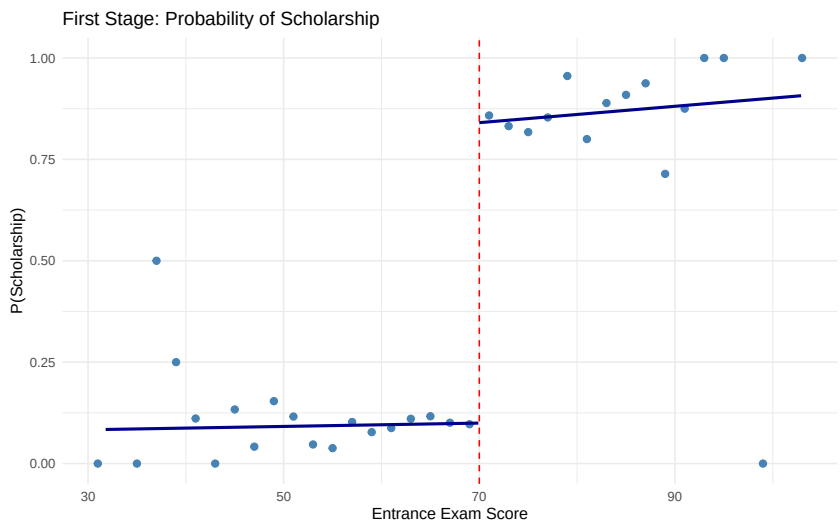
Result: As predicted, the optimal bandwidth is wider on the right where observations are fewer. The asymmetric bandwidth selection adapts to the data distribution.

Exercise D: Fuzzy RDD

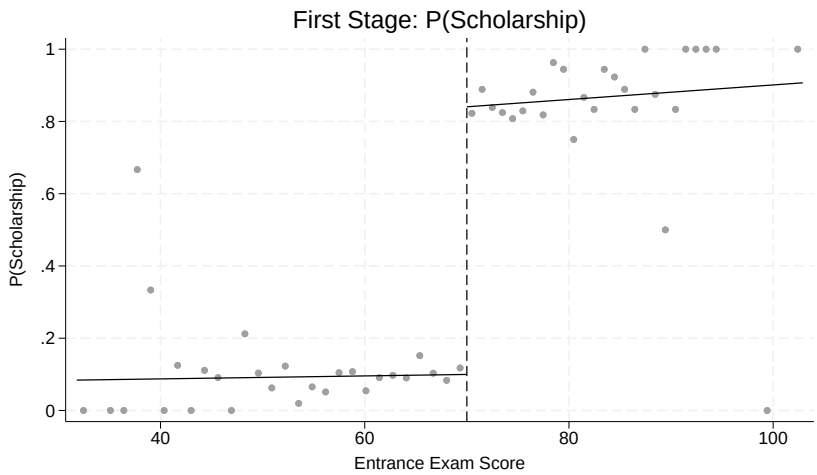
D.1 *(0.5 pts)* First stage plot and estimate.

Solution:

First Stage Plot:
R Output:



Stata Output:



First Stage Estimate:

Jump in P(scholarship) at cutoff	$\boxed{0.708}$
Standard error	0.037

Interpretation: Crossing the cutoff increases the probability of receiving a scholarship by about 71 percentage points. This is less than 1 because:

- Some students above decline ($\approx$ 15% never-takers)
- Some students below receive it through appeals ($\approx$ 10% always-takers)

D.2 *(0.5 pts)* Reduced form estimate.

Solution:

Reduced Form Estimate:

Jump in GPA at cutoff	$\boxed{0.335}$
Standard error	0.038

This is the **intent-to-treat (ITT)** effect: the effect of being *eligible* for a scholarship (crossing the threshold), not necessarily receiving one.

D.3 *(1 pt)* Fuzzy RDD estimate and 2SLS verification.

Solution:

Wald Estimate:

$$\hat{\tau}_{Fuzzy} = \frac{\hat{\rho}}{\hat{\pi}_1} = \frac{0.335}{0.708} = \boxed{0.473}$$

2SLS Verification:
Using $\mathbb{1}(X_i \geq 70)$ as instrument for scholarship:

	Estimate	Std. Error	t-value	p-value
Scholarship (instrumented)	0.473	0.047	10.02	< 0.001

The 2SLS estimate matches the Wald estimate, confirming the equivalence.
rdrobust (fuzzy = TRUE): $\hat{\tau} = 0.530$ (SE = 0.066)
The rdrobust estimate is slightly higher due to different bandwidth and bias correction.

D.4 *(1 pt)* LATE interpretation.

Solution:

(a) Who are the compliers, always-takers, and never-takers?

Type	Description	Proportion
Compliers	Students who receive scholarship if and only if above cutoff. They accept when eligible and don't appeal when ineligible.	$\approx 75\%$
Always-takers	Students who receive scholarship regardless of score. They appeal successfully when below cutoff.	$\approx 10\%$
Never-takers	Students who don't receive scholarship regardless of score. They decline when above cutoff (e.g., receive a better offer).	$\approx 15\%$

(b) Why fuzzy RDD might differ from sharp RDD:

1. **Different estimands:**
   - Sharp RDD: ATE for all students near cutoff

- Fuzzy RDD: LATE for compliers only

2. **Complier heterogeneity:** Compliers are "marginal" students—those whose scholarship decision depends on the cutoff. They may have different treatment effects than always-takers (who sought scholarships aggressively) or never-takers (who didn't value the scholarship).

3. **Selection:** If students who appeal (always-takers) are more motivated and would benefit less from the scholarship, or if students who decline (never-takers) have better outside options, the complier effect could differ from the population effect.

Note: In this simulation, the true effect for compliers is 0.5 GPA points. Our estimate of 0.47–0.53 is close to the truth.